# Constrained Linear

## Parametrization

This model is like a "fixed" effect where you can constrained the coefficient of a covariate to be in an interval:

$$\eta_i = \beta x_i$$

where $\beta$ is in the interval [`low`, `high`] and $x$ are the covariates.

## Hyperparameters

The $\beta$ parameter, since its is constrained in general, is a hyperparamter. The internal transformation depends on the values of `low` and `high`. If `low` is `-Inf` and `high` is `Inf`, then

$$\beta = \theta$$

and the prior is put on $\theta$. If `low` is finite and `high` is `Inf`, then

$$\beta = \text{low} + \exp(\theta)$$

and the prior is put on $\theta$. If `low` is finite and `high` is finite, then

$$\beta = \text{low} + (\text{high} - \text{low})\frac{\exp(\theta)}{1 + \exp(\theta)}$$

and the prior is put on $\theta$.

## Specification

```
f(x, model="clinear", range = c(low, high), precision = <precision>)
```

where `precision` is the precision for the tiny noise used to implement this as a latent model.

### Hyperparameter spesification and default values

**doc** `Constrained linear effect`

**hyper**

**theta**

**hyperid** `37001`
**name** `beta`
**short.name** `b`
**initial** `1`
**fixed** `FALSE`
**prior** `normal`
**param** `1 10`
**to.theta** `function(x, REPLACE.ME.low, REPLACE.ME.high) {`
**from.theta** `function(x, REPLACE.ME.low, REPLACE.ME.high) {`

**constr** `FALSE`

**nrow.ncol** `FALSE`

**augmented** `FALSE`

**aug.factor** `1`

**aug.constr**

**n.div.by**

**n.required** `FALSE`

**set.default.values** `FALSE`

**pdf** `clinear`

## Example

```
n = 100
x = runif(n)
y = 1 + x + rnorm(n)
r = inla(y ~ f(x, model = "clinear", range = c(0, Inf)),
         data = data.frame(y,x))
summary(r)
```

## Notes

None